

Cutoff Theorems for the Equivalence of Parameterized Quantum Circuits

Neil J. Ross and **Scott Wesley**

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DALHOUSIE
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Quantum Computing Background

What is a Quantum Circuit

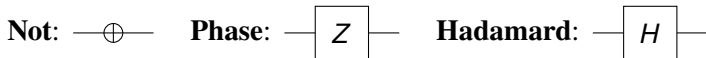
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Objectives

Extending the
Miller-Bakewell
Technique

Conclusions

In this talk, we will think of quantum circuits as mathematical objects. A **quantum circuit** is composed from gates.



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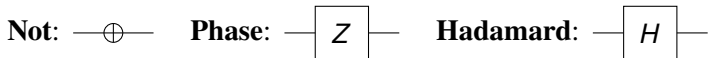
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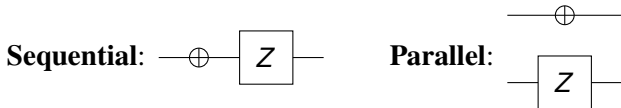
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Gates can be composed sequentially and in parallel.



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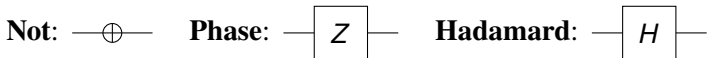
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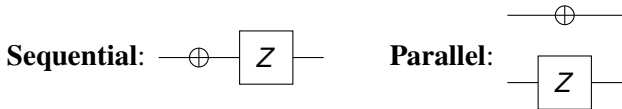
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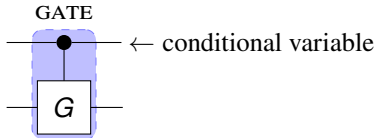
In this talk, we will think of quantum circuits as mathematical objects. A **quantum circuit** is composed from gates.



Gates can be composed sequentially and in parallel.



Gates can also be controlled, giving rise to conditional.



Quantum Circuit Semantics

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The interpretation of a gate is a unitary matrix.

$$\llbracket \text{---}\oplus\text{---} \rrbracket = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \llbracket \text{---}\boxed{Z}\text{---} \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \llbracket \text{---}\boxed{H}\text{---} \rrbracket = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

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Controls are defined inductively via block decompositions.

$$\llbracket \begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \boxed{G} \text{---} \end{array} \rrbracket = \begin{bmatrix} I & 0 \\ 0 & \llbracket G \rrbracket \end{bmatrix}$$

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Sequential and parallel composition are matrix multiplication and the Kronecker tensor product.

$$\llbracket -\boxed{C_1}-\boxed{C_2}- \rrbracket = \llbracket C_2 \rrbracket \llbracket C_1 \rrbracket \quad \llbracket \begin{array}{c} \boxed{C_1} \\ \boxed{C_2} \end{array} \rrbracket = \llbracket C_1 \rrbracket \otimes \llbracket C_2 \rrbracket$$

The Kronecker Tensor Product

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$$\begin{bmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,n} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m,1} & c_{m,2} & \cdots & c_{m,n} \end{bmatrix} \otimes M = \begin{bmatrix} c_{1,1}M & c_{1,2}M & \cdots & c_{1,n}M \\ c_{2,1}M & c_{2,2}M & \cdots & c_{2,n}M \\ \vdots & \vdots & \ddots & \vdots \\ c_{m,1}M & c_{m,2}M & \cdots & c_{m,n}M \end{bmatrix}$$

Parameterized Quantum Circuits (PQCs)

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If M is Hermitian and unitary, then the following is true.

$$\exp(iM\theta) = \sum_{n=0}^{\infty} \frac{(iM\theta)^n}{n!} = \cos(\theta)I + i \sin(\theta)M$$

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Given a Hermitian unitary M , we define a rotation gate as follows.

$$\left[\text{---} \boxed{R_M(\theta)} \text{---} \right] = \exp(-iM\theta/2)$$

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A parameterized quantum circuit (PQC) is a quantum circuit using parameterized gates. These circuits appear in theoretical computer science, algorithm design, and program optimization.

Cyclotomic Numbers and Gate Sets

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In principle, any square unitary matrix could be a quantum gate.
However, the entries of most unitary matrices are not computable.

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In practice, approximately universal finite gate sets are used.

Clifford (non-universal)	Clifford+T	Clifford+ $\sqrt{T}$
Toffoli+Hadamard	Clifford-Cyclotomic	

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Cyclotomic Field Extensions. A cyclotomic extension of $\mathbb{Q}$ is a field $\mathbb{Q}(\zeta)$ where ζ is some root of unity. All gate sets listed above are definable over cyclotomic extensions of $\mathbb{Q}$.

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Efficient Computation. All cyclotomic extensions embed into the universal cyclotomic field. Arithmetic can be computed *efficiently* over this field.

Laurent Polynomials and Their Degrees

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Multivariate Laurent polynomials generalize multivariate polynomials to allow for negative degrees.

$$f(x) = x^{-5} - 3x^2 + x - 10 \quad g(x, y) = xy^{-3} + x^2y - x^{-1}y + 7$$

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Multivariate Laurent polynomials have both positive and negative degrees associated to each variable.

$$\deg_x^+(f) = 2 \quad \deg_x^-(f) = 5 \quad \deg_x^+(g) = 2 \quad \deg_x^-(g) = 1$$

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Combinatorial Nullstellensatz (Extended). Let $f \neq 0$ be a polynomial over a field $\mathbb{F}$. If $S_1, \dots, S_k$ are subsets of $\mathbb{F}$ with $|S_j| > \deg_{x_j}^-(f) + \deg_{x_j}^+(f)$ for each j , then there exists $x \in S_1 \times S_2 \times \dots \times S_k$ such that $f(x) \neq 0$.

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Remark. If $f(x) = 0$ for all $x \in S_1 \times S_2 \times \dots \times S_k$, then f is identically zero.

Motivation and Objectives

Parameterized Equivalence Checking

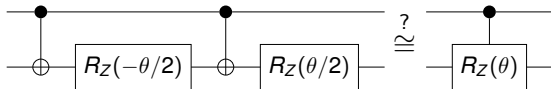
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Parameterized equivalence checking (PEC) asks where two PQCs are equivalence (in some sense of equivalence).



Parameterized Equivalence Checking

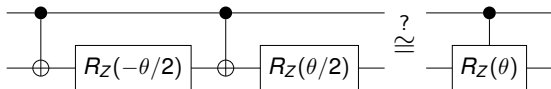
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Parameterized equivalence checking (PEC) asks where two PQC's are equivalence (in some sense of equivalence).



There are many notions of equivalence for quantum circuits.

Exact	Modulo-Noise	Modulo-Constant-Phase
Modulo-Phase	Modulo-Failure	Observable

Decidability for the Parameter-Free Case

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Let $\mathcal{G}$ be a parameter-free gate set whose matrices have entries over a computable field.

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Let $\mathcal{G}$ be a parameter-free gate set whose matrices have entries over a computable field.

If C is a circuit over $\mathcal{G}$ then entry (j, k) of $\llbracket C \rrbracket$ as follows.

$$\begin{aligned} \left[\begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \boxed{A} \text{---} \end{array} \right]_{j,k} &= \begin{cases} I_{j,k} & \text{if } j, k < n \\ \llbracket A \rrbracket_{j-n, k-n} & \text{if } j, k \geq n \\ 0 & \text{otherwise} \end{cases} \\ \left[\begin{array}{c} \text{---} \boxed{A} \text{---} \\ \text{---} \boxed{B} \text{---} \end{array} \right]_{j,k} &= \llbracket A \rrbracket_{f(j,k)} \llbracket B \rrbracket_{g(j,k)} \\ \left[\text{---} \boxed{A} \text{---} \boxed{B} \text{---} \right]_{j,k} &= \sum_{\ell=1}^n \llbracket B \rrbracket_{j,\ell} \llbracket A \rrbracket_{\ell,k} \end{aligned}$$

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Let $\mathcal{G}$ be a parameter-free gate set whose matrices have entries over a computable field.

If $\mathcal{C}$ is a circuit over $\mathcal{G}$ then entry (j, k) of $\llbracket \mathcal{C} \rrbracket$ as follows.

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This gives a way to check whether $\llbracket \mathcal{C}_1 \rrbracket = \llbracket \mathcal{C}_2 \rrbracket$.

Progress Towards Exact PEC

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Quartz (2022). A decision procedure for an ad-hoc gate set by reduction to satisfiability modulo non-linear real arithmetic. The complexity of the black-box is worse-case double exponential.

$$\llbracket -\boxed{R_X(\theta)}- \rrbracket = \begin{bmatrix} \cos(\theta) & i \sin(\theta) \\ i \sin(\theta) & \cos(\theta) \end{bmatrix} = \begin{bmatrix} x & iy \\ iy & x \end{bmatrix} \text{ where } x^2 + y^2 = 1$$

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QCEC (2023). A randomized algorithm that *almost surely* succeeds for *analytic* rotations via continuous uniform sampling.



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QCEC (2023). A randomized algorithm that *almost surely* succeeds for *analytic* rotations via continuous uniform sampling.



Queso (2023). A randomized algorithm with discrete sampling based on the Schwartz-Zippel lemma. Determining the sample size requires expanding out the matrix.



Early Cutoff Results (Miller-Bakewell – 2019)

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The Miller-Bakewell technique provides a solution to exact PEC for gate sets expressed over the ZX-, ZH-, or ZW-calculus.

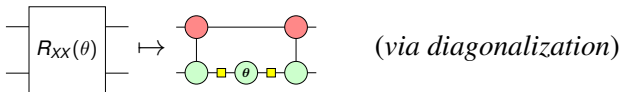
Step 1. Find a change over variable under which each matrix can be written as a complex Laurant polynomial.

Step 2. Compute the multivariate degree of $\llbracket C \rrbracket$ using rules such that $\deg_Y \llbracket C_1 \otimes C_2 \rrbracket \leq \deg_Y \llbracket C_1 \rrbracket + \deg_Y \llbracket C_2 \rrbracket$.

Step 3. Reduce to PEC finitely many parameter-free instances through linear interpolation (*or combinatorial Nullstellensatz*).

Limitations of the Miller-Bakewell Technique

- Writing arbitrary rotations in ZX-calculus is challenging.



Limitations of the Miller-Bakewell Technique

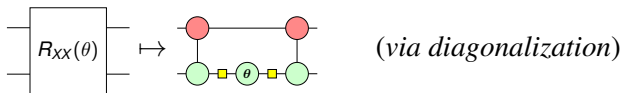
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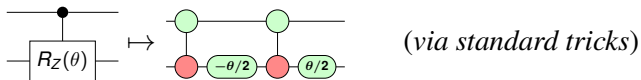
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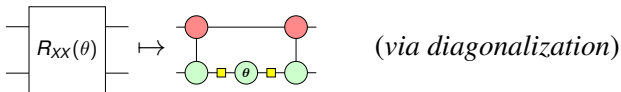


- Writing controls in ZX-calculus is challenging.

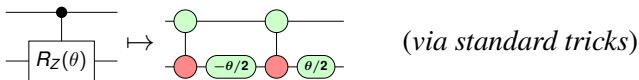


Limitations of the Miller-Bakewell Technique

- Writing arbitrary rotations in ZX-calculus is challenging.



- Writing controls in ZX-calculus is challenging.



- In the ZX-case, parameters are assumed to be single angles.

$$\llbracket -\alpha+\beta \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{j(\alpha+\beta)} \end{bmatrix} = \boxed{???} \quad (\text{no rules given})$$

- The degree bounds are not closed form equations.

The Challenge of Affine Linear Global Phase

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There are many notions of global phase for PQCs.

The Challenge of Affine Linear Global Phase

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There are many notions of global phase for PQCs.

Parameterized Global Phase. Two parameterized circuits are equivalent modulo global phase if for each parameter $\mathbf{x} \in \mathbb{R}^k$, there exists a $\mathbf{c} \in \mathbb{C}$ with $|\mathbf{c}| = 1$ such that $\llbracket C_1 \rrbracket(\mathbf{x}) = \mathbf{c} \llbracket C_2 \rrbracket(\mathbf{x})$.

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Affine Linear Global Phase. Two circuits are equivalent modulo affine linear global phase if there exists coefficients $\alpha \in \mathbb{R}^k$ and $\beta \in \mathbb{R}$ satisfying the following equation.

$$\llbracket C_1 \rrbracket(\theta) = e^{j(\alpha_1 \theta_1 + \dots + \alpha_k \theta_k + \beta)} \llbracket C_2 \rrbracket(\theta)$$

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$$\llbracket C_1 \rrbracket(\theta) = e^{j(\alpha_1 \theta_1 + \dots + \alpha_k \theta_k + \beta)} \llbracket C_2 \rrbracket(\theta)$$

Quartz and QCEC give semi-decision procedures for reducing PEC modulo affine linear global phase to exact PEC.

Towards a Circuit-Based Approach

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This paper can be seen as generalizing the Miller-Bakewell technique in several directions.

Target Problem. PEC modulo affine linear global phase.

Target Circuits. Arbitrary parameterized gate sets with sequential composition, parallel composition, and control modifiers.

Characterizing the Bounds. We wish to find closed form bounds computed from the syntactic structure of the circuits.

Characterizing the Circuits. We further aim to characterize a (large) class of quantum gates for which this method is applicable.

Extending the Miller-Bakewell Technique

Definition: Integral Circuit

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Recall the change of variable in the Miller-Bakewell technique.

$$\llbracket -\textcircled{\theta} \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Y \end{bmatrix} \text{ where } Y = e^{i\theta}$$

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It is easy to generalize this to integer coefficients.

$$\llbracket -\textcircled{2\theta} \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{i2\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Y^2 \end{bmatrix}$$

Definition: Integral Circuit

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It is easy to generalize this to integer coefficients.

$$\llbracket -\textcircled{2\theta} - \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{j2\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \gamma^2 \end{bmatrix}$$

However, rational coefficients are more problematic.

$$\llbracket -\textcolor{lightgreen}{\theta/3} - \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{j\theta/3} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \gamma^{1/3} \end{bmatrix}$$

Definition: Integral Circuit

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Recall the change of variable in the Miller-Bakewell technique.

$$\llbracket -\textcircled{\theta} \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Y \end{bmatrix} \text{ where } Y = e^{i\theta}$$

It is easy to generalize this to integer coefficients.

$$\llbracket -\textcircled{2\theta} \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{i2\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Y^2 \end{bmatrix}$$

However, rational coefficients are more problematic.

$$\llbracket -\textcircled{\theta/3} \rrbracket = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta/3} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Y^{1/3} \end{bmatrix}$$

We will say that a circuit is integral if all parameters are affine linear variable combinations with integer coefficients.

$$\text{---} \boxed{R_Z(\theta)} \text{---} \quad \text{---} \boxed{R_Z(5\theta)} \text{---} \quad \text{---} \boxed{R_X(3\alpha + 7\beta - 2\gamma + \pi/3)} \text{---}$$

A Reduction to Polynomial Equality

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Recall the following trigonometric identities.

$$\cos(x) = \frac{1}{2} (e^{ix} + e^{-ix}) \qquad \sin(x) = \frac{1}{2i} (e^{ix} - e^{-ix})$$

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When applied to $\cos(-f(\theta)/2)$, the following is obtained.

$$\cos(-f(\theta)/2) = \frac{1}{2} \left(e^{-ib/2} \prod_{j=1}^k e^{-ia_j \theta_j / 2} + e^{ib/2} \prod_{j=1}^k e^{ia_j \theta_k / 2} \right)$$

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Taking $z_j = e^{-i\theta_j/2}$, the following Laurent polynomial is obtained.

$$\cos(-f(\theta)/2) = \left(\frac{e^{-ib/2}}{2} \left(\prod_{j=1}^k z_j^{a_j} + \left(\frac{e^{ib/2}}{2} \right) \prod_{j=1}^k z_j^{-a_j} \right) \right)$$

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Then $\llbracket R_M(f(\theta)) \rrbracket = \cos(-f(\theta)/2)I + i \sin(-f(\theta)/2)M$ can be written as a matrix of Laurent polynomials.

Decidability for Cyclotomic Circuits

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Polynomial Semantics. By the change of variable $z_j = e^{-i\theta_j/2}$, each $\llbracket C \rrbracket(\theta)$ is a matrix of Laurent polynomials over $(z_1, \dots, z_k)$.

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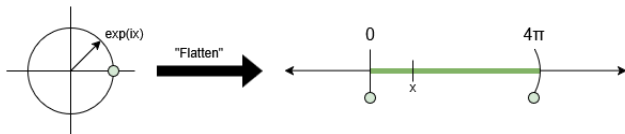
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Evaluation. Given $x \in \mathbb{R}^k$, evaluating $\llbracket C \rrbracket$ at x corresponds to evaluating the polynomial at $(e^{-ix_1/2}, \dots, e^{-ix_k/2})$. We can evaluate the polynomial at unique points by sampling $[0, 4\pi)$.



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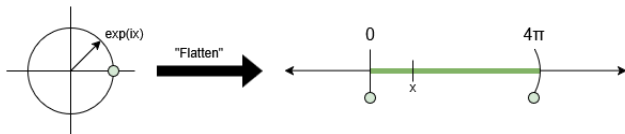
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Reduction. Combinatorial Nullstellensatz for $\llbracket C_2 \rrbracket - \llbracket C_1 \rrbracket = 0$.

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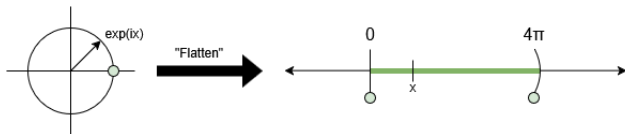
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Reduction. Combinatorial Nullstellensatz for $\llbracket C_2 \rrbracket - \llbracket C_1 \rrbracket = 0$.

Decidability. If $\theta \in \mathbb{Q}\pi$, then $e^{i\theta/2}$ is cyclotomic. By sampling from $([0, 4\pi) \cap \mathbb{Q}\pi)^k$, we obtain decidability for cyclotomic gates.

Closed Form Equation for a Coarse Degree Bound

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Define the coefficient sequence of a rotation gate as follows.

$$\boxed{R_Z(a_1\theta_1 + \dots + a_k\theta_k + b)} \mapsto (a_1, \dots, a_k)$$

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Let $A(C)$ denote the list of each coefficient sequence of each rotation gate in C . It follows by a careful inductive argument that these coefficients bound the degree: $\deg_{Z_j}^{\pm}(\llbracket C \rrbracket) \leq \sum_{a \in A(C)} |a_j|$.

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Necessity. For each coefficient sequence α , there exists a circuit C with $A(c) = \alpha$ for which the bound is tight.

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Coarseness. For each $n \in \mathbb{N}$ there exists a single qubit circuit C such that $\deg_{Z_j}^{\pm}(\llbracket C \rrbracket) = 0$ but $\sum_{a \in A(C)} |a_j| = 2n$.

$$\boxed{R_X(n\theta)} \boxed{Z} \boxed{R_X(n\theta)} \cong \boxed{Z}$$

From Degree Bounds to Global Phases

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Let $f(\theta) = \alpha_1\theta_1 + \cdots + \alpha_k\theta_k + \beta$ and $\lambda_j = \sum_{a \in A(C)} |a_j|$. We want to check if $\llbracket C_1 \rrbracket = e^{if(\theta)} \llbracket C_2 \rrbracket$ for some α and β .

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A Simple Idea. Since $\llbracket C_1 \rrbracket$ is a polynomial, we should be able to solve for α and β via a modified version of linear interpolation.

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The Roadblock. We do not know a-priori whether α_j is rational. Moreover, $e^{i\alpha_j\theta_j}$ is only injective on $[0, 4\pi/\alpha_j)$, so we need to know the magnitude of α_j .

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Key Insight. The term $e^{i\alpha_j\theta_j}$ contributes to the degree of $\llbracket C_1 \rrbracket$, so we can show it is rational and bounded by $2\lambda_j$. Otherwise, the α_j would violate the degree bound on $\llbracket C_1 \rrbracket$!

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Summary of Results and Future Work

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In this talk, we saw that parameterized equivalence checking modulo affine linear global phase is decidable. There are several directions for future work.

- **Wider Circuit Classes.** Can this technique be extended to gates that are not isometries (e.g., post-selection)?
- **Equivalence Relations.** Can this technique be extended to other notions of equivalence?
- **Complexity.** What class does this problem belong to? What if the number of wires or parameters are restricted?
- **Symmetries.** Can symmetries between parameters be used to reduce the number of instances required for PEC.
- **Data Structures.** Which data structures from parameter-free equivalence checking can be extended to support sampling?